Bangladesh University of Engineering and Technology
Department of Computer Science & Engineering

MATH 241 Question Bank

Vector Calculus · Complex Calculus · PDE

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BUET · MATH 241 · Advance Calculus

Vector Calculus

Vector Algebra, Gradient, Divergence, Curl, Integration Theorems

S1. Vector Calculus

Vector Algebra

Q1. Examine whether the vectors $3\mathbf{i}+2\mathbf{j}-\mathbf{k}$, $-\mathbf{i}+4\mathbf{j}-5\mathbf{k}$ and $-2\mathbf{i}+\mathbf{j}-3\mathbf{k}$ form a linearly dependent or independent set. If dependent, find a linear relation among them. **(15 marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q2. Find a set of vectors reciprocal to: $2\mathbf{i}+3\mathbf{j}-\mathbf{k}$, $\mathbf{i}-\mathbf{j}-2\mathbf{k}$, $-\mathbf{i}+2\mathbf{j}+2\mathbf{k}$. **(22 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q3. Show that:

(a) $(\mathbf{a}\times\mathbf{b})\cdot(\mathbf{b}\times\mathbf{c})\times(\mathbf{c}\times\mathbf{a}) = (\mathbf{a}\cdot\mathbf{b}\times\mathbf{c})^2$

(b) $(\mathbf{a}\times\mathbf{b})\times(\mathbf{c}\times\mathbf{d}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}\times\mathbf{d}) - \mathbf{a}(\mathbf{b}\cdot\mathbf{c}\times\mathbf{d})$

(30 marks) [MATH 147 | L-1/T-2 | 2018–19]

Q4. Examine whether $\mathbf{u}=3\mathbf{i}+2\mathbf{j}-\mathbf{k}$, $\mathbf{v}=-\mathbf{i}+4\mathbf{j}-5\mathbf{k}$, $\mathbf{w}=-2\mathbf{i}+\mathbf{j}-3\mathbf{k}$ are linearly dependent or independent. If $\mathbf{x}=-\mathbf{i}-3\mathbf{j}+6\mathbf{k}$, find a,b,c such that $\mathbf{x}=a\mathbf{u}+b\mathbf{v}+c\mathbf{w}$. **(23 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q5. Given $\mathbf{p}=3\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{q}=\mathbf{i}-2\mathbf{j}-4\mathbf{k}$ are position vectors of P and Q.

(a) Find an equation for the plane passing through Q perpendicular to line PQ.

(b) What is the distance from $(-1,1,1)$ to the plane?

(23 marks) [MATH 147 | L-1/T-2 | 2017–18]

Q6. Prove that the vectors $\mathbf{A}=3\mathbf{i}+\mathbf{j}-2\mathbf{k}$, $\mathbf{B}=-\mathbf{i}+3\mathbf{j}+4\mathbf{k}$, $\mathbf{C}=4\mathbf{i}-2\mathbf{j}-6\mathbf{k}$ can form the sides of a triangle. Find the lengths of the medians of the triangle. **(20 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Q7. Let $\mathbf{a}=\mathbf{i}-3\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=2\mathbf{i}-4\mathbf{j}-\mathbf{k}$. Find the vector component of $\mathbf{a}$ in the direction of $\mathbf{b}$ and perpendicular to $\mathbf{b}$. **(13 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Q8. Show that any vector $\mathbf{r}$ can be represented as a linear combination of three non-coplanar vectors $\mathbf{a},\mathbf{b},\mathbf{c}$. Hence find a linear relation among $(2,-3,4)$, $(1,-1,1)$, $(-1,1,1)$ and $(1,1,1)$. **(20 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q9. Prove that $[\mathbf{a}\times\mathbf{p},\ \mathbf{b}\times\mathbf{q},\ \mathbf{c}\times\mathbf{r}]+[\mathbf{a}\times\mathbf{q},\ \mathbf{b}\times\mathbf{r},\ \mathbf{c}\times\mathbf{p}]+[\mathbf{a}\times\mathbf{r},\ \mathbf{b}\times\mathbf{p},\ \mathbf{c}\times\mathbf{q}]=0$. **(13 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Differentiation and Integration of Vectors

Q1. Define osculating plane. Find the curvature of a circular helix at any point on it. Comment on the result. **(12 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q2. A particle moves through 3-space so that its position vector at time t is $\mathbf{r}=t\,\hat{\mathbf{i}}+t^2\,\hat{\mathbf{j}}+t^3\,\hat{\mathbf{k}}$. Find the vector tangential and normal components of acceleration at $t=1$. **(15 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q3. Find the unit tangent $\mathbf{T}$, principal normal $\mathbf{N}$ and binormal $\mathbf{B}$ for the space curve $x=t-t^3/3$, $y=t^2$, $z=t+t^3/3$. **(24 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q4. Find the unit tangent at the point where $t=2$ on the curve $x=t^2+1$, $y=4t-3$, $z=2t^2-6t$. **(10 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q5. A particle moves along $x=2t^2$, $y=t^2-4t$, $z=3t-5$. Find the components of velocity and acceleration at $t=1$ in the direction $\mathbf{i}-3\mathbf{j}+2\mathbf{k}$. **(14 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Gradient and Directional Derivative

Q1. Find equations of the tangent plane and normal line to the surface $xz-yz^3+yz^2=2$ at $(2,-1,1)$. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Find the directional derivative of $f(x,y,z)=y-\sqrt{x^2+z^2}$ at $(-3,1,4)$ in the direction of $2\mathbf{i}-2\mathbf{j}-\mathbf{k}$. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q3. If $\nabla\phi = 2xyz^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$, find $\phi(x, y, z)$ when $\phi(1, -2, 2) = 4$. **(13 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q4. Write down some applications of gradient. Find the angle between the surfaces $xy^2z - 3x - z^2 = 0$ and $3x^2 - y^2 + 2z = 1$ at the point $(1, -1, 2)$. **(8 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q5. Find equations for the tangent plane and normal line to the surface $4z = x^2 - y^2$ at $(3, 1, 2)$. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q6. Find the directional derivative of $\phi(x, y, z) = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction of $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q7. Find an equation of the tangent plane to the ellipsoid $x^2 + 4y^2 + z^2 = 18$ at $(1, 2, 1)$. **(15 marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q8. Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction of $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$. **($16\frac{2}{3}$ marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q9. Find the values of constants a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at $(1, 2, -1)$ has a maximum magnitude 64 in the direction parallel to the z-axis. **(15 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q10. In what direction from $(2, 1, -1)$ is the directional derivative of $\phi = x^2yz^3$ maximum? What is the maximum value of the directional derivative? **(14 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Q11. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = |\mathbf{r}|$, prove that $\operatorname{div}\{\frac{f(r)\mathbf{r}}{r}\} = \frac{1}{r^2}\frac{d}{dr}\{r^2 f(r)\}$. **(13 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q12. Find the equation of the binormal at any point t of the curve $x = 3\cos t$, $y = 3\sin t$, $z = 4t$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q13. Find the values of constants a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^3$ at $(1, 2, -1)$ has a maximum magnitude 64 in the direction parallel to the z-axis. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Divergence and Curl

Q1. Show that (i) $\nabla r^n = nr^{n-2}\mathbf{r}$, (ii) $\phi = \frac{1}{r}$ is a solution of $\nabla^2\phi = 0$. **(15 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Define a solenoidal vector field. Find the most general function $f(r)$ so that $f(r)\,\mathbf{r}$ is solenoidal. **(10 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q3. Test whether $\mathbf{E} = \frac{\mathbf{r}}{r^2}$ is irrotational. If so, find ϕ such that $\mathbf{E} = \operatorname{grad}\phi$ with $\phi(a) = 0$, $a > 0$. **(12 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q4. Show that $\mathbf{F} = (6xy + z^3)\mathbf{i} + (3x^2 - z)\mathbf{j} + (3xz^2 - y)\mathbf{k}$ is irrotational. Hence find ϕ such that $\mathbf{F} = \nabla\phi$. **(13 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q5. Prove that $\nabla \times (\nabla \times \mathbf{F}) = -\nabla^2\mathbf{F} + \nabla(\nabla \cdot \mathbf{F})$. **(15 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q6. Prove that $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}$. **(13 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Vector Calculus Identities: Jacobian, Hessian, Laplacian

Q1. Evaluate $\iint_R e^{xy}\,dA$ where R is the region enclosed by $y = \frac{1}{2}x$, $y = x$, $y = \frac{1}{x}$ and $y = \frac{2}{x}$. **(12 marks)** [MATH 241 | L-2/T-1 | 2024–25, 2023–24]

Q2. Define Jacobian matrix, Hessian matrix and Laplacian operator. **(5 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q3. Evaluate $\iint_R \frac{x - y}{x + y}\,dA$ where R is the region enclosed by $x - y = 0$, $x - y = 1$, $x + y = 1$ and $x + y = 3$. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Line Integrals

Q1. Find the work done in moving a particle in $\mathbf{F}(x, y, z) = (3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$ along the straight lines from $(1, 0, 0)$ to $(1, 1, 0)$ and then to $(1, 1, 1)$. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. State Green's theorem and verify in the plane for $\oint_C (3x^2 - 8y^2)\, dx + (4y - 6xy)\, dy$ where C bounds the region enclosed by $y = \sqrt{x}$ and $y = x^2$. **(12 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q3. Find the work done by $\mathbf{F} = (3x^2 + 2y)\hat{\mathbf{i}} - (x + 3\cos y)\hat{\mathbf{j}}$ around the parallelogram with vertices $(0, 0)$, $(2, 0)$, $(3, 1)$, $(1, 1)$. **(13 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q4. Find the work done in moving a particle in $\mathbf{F}(x, y, z) = 3x^2\mathbf{i} + (2xz - y)\mathbf{j} + z\mathbf{k}$ along $x^2 = 4y$, $3x^3 = 8z$ from $x = 0$ to $x = 2$. **(10/15/22 marks)** [MATH 241 | L-2/T-1 | 2022–23] [MATH 147 | L-1/T-2 | 2020–21, 2019–20]

Q5. State Green's theorem. Verify it for $\oint_C (3x^2 - 8y^2)\, dx + (4y - 6xy)\, dy$ where C bounds the region enclosed by $y = x^2$ and $x = y^2$. **(12 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q6. Evaluate $\oint_C [(x + 2y)\, dx + (x - y)\, dy]$ along C: $x = 2\cos t$, $y = 4\sin t$, $0 \le t \le \pi/4$. **(16 marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q7. Use Green's Theorem to evaluate $\oint_C x^2y\, dx - xy^2\, dy$ where C bounds the first-quadrant region enclosed by the coordinate axes and the circle $x^2 + y^2 = 16$. **($15\frac{1}{2}$ marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q8. Using Stokes' theorem evaluate $\oint_C(\sin z\, dx - \cos x\, dy + \sin y\, dz)$, where C is the boundary of the rectangle $0 \le x \le \pi$, $0 \le y \le 1$, $z = 3$. **(22 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q9. If $\mathbf{A} = (3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$, evaluate $\int_C \mathbf{A} \cdot d\mathbf{r}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along straight lines $(0, 0, 0) \to (1, 0, 0) \to (1, 1, 0) \to (1, 1, 1)$. **(16 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q10. Use a line integral to find the area of the ellipse $x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$. **(16 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q11. If $\mathbf{A} = (2y + 3)\mathbf{i} + xz\mathbf{j} + (yz - x)\mathbf{k}$, evaluate $\int_C \mathbf{A} \cdot d\mathbf{r}$ along straight lines from $(0, 0, 0)$ to $(1, 0, 0)$, then to $(1, 1, 0)$, then to $(2, 1, 1)$. **(16 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Q12. Use a line integral to find the area of the region enclosed by the asteroid $x = a\cos^3\phi$, $y = a\sin^3\phi$, $0 \le \phi \le 2\pi$. **(16 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Q13. If $\mathbf{F} = (y^2\cos x + z^3)\mathbf{i} + (2y\sin x - 4)\mathbf{j} + (3xz^2 + 2)\mathbf{k}$, show that $\int \mathbf{F} \cdot d\mathbf{r}$ is independent of path. Find the work done in moving an object from $(0, 1, -1)$ to $(\pi/2, -1, 2)$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Surface Integrals & Divergence Theorem

Q1. Evaluate $\iint_S \mathbf{F} \cdot \hat{n}\, dS$ where $\mathbf{F} = (x + y^2)\mathbf{i} - 2x\mathbf{j} + 2yz\mathbf{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant. **(20 marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q2. If $\mathbf{F} = (2x^2 - 3z)\mathbf{i} - 2xy\mathbf{j} - 4x\mathbf{k}$, evaluate $\iiint_V \nabla \cdot \mathbf{F}\, dV$ where V is the closed region bounded by $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$. **(24 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q3. Using the Divergence Theorem evaluate $\iint_S \mathbf{F} \cdot \hat{n}\, dS$, where $\mathbf{F} = 2xy\mathbf{i} + yz^2\mathbf{j} + xz\mathbf{k}$ and S is the surface of the parallelepiped bounded by $x = 0$, $y = 0$, $z = 0$, $x = 2$, $y = 1$ and $z = 3$. **(24 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q4. Evaluate $\iiint_V \nabla \cdot \mathbf{F}\, dV$ where $\mathbf{F} = yx^2\mathbf{i} + (xy^2 - 3z^4)\mathbf{j} + (x^3 + y^3)\mathbf{k}$ and V is the region bounded by $x^2 + y^2 + z^2 = 16$, $y \le 0$ and $z \ge 0$. **(25 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q5. Evaluate $\iint_S \mathbf{A} \cdot \hat{n}\, dS$, where $\mathbf{A} = 18z\mathbf{i} - 12\mathbf{j} + 3y\mathbf{k}$ and S is that part of the plane $2x + 3y + 6z = 12$ in the first octant. **(23 marks)** [MATH 147 | L-1/T-2 | 2017–18, 2016–17]

Q6. Use the Divergence Theorem to find the outward flux of $\mathbf{F} = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$ across the surface of the region enclosed by $z = \sqrt{a^2 - x^2 - y^2}$ and the plane $z = 0$. **(23 marks)** [MATH 147 | L-1/T-2 | 2017–18, 2016–17]

Q7. State the Divergence Theorem and verify it for $\mathbf{F} = (x + y^2)\mathbf{i} - 2x\mathbf{j} + 2yz\mathbf{k}$ for the tetrahedron bounded by the coordinate planes and the plane $2x + y + 2z = 6$. **(16 marks)** **[MATH 147 | L-1/T-2 | 2015–16]**

Stokes' Theorem

Q1. Verify Stokes' Theorem for $\mathbf{F}(x, y, z) = 2z\mathbf{i} + x\mathbf{j} + y^2\mathbf{k}$ where S is the surface of the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane, oriented upward, and C is its boundary in the xy-plane, oriented counterclockwise. **($26\frac{2}{3}$ marks)** **[MATH 147 | L-1/T-2 | 2020–21]**

Q2. State Stokes' theorem and verify it for $\mathbf{F} = (z^2 - 1)\mathbf{i} + (z + xy^3)\mathbf{j} + 6\mathbf{k}$ taken over the portion of $x = 6 - 4y^2 - 4z^2$ in front of $x = -2$. **(40 marks)** **[MATH 147 | L-1/T-2 | 2018–19]**

Q3. Evaluate $\iint (\nabla \times \mathbf{A}) \cdot \hat{n}\, dS$ where $\mathbf{A} = (x^2 + y - 4)\mathbf{i} + 3xy\mathbf{j} + (2xz + z^2)\mathbf{k}$ and S is the surface of the hemisphere $x^2 + y^2 + z^2 = 16$ above the xy-plane. **(26 marks)** **[MATH 147 | L-1/T-2 | 2015–16]**

BUET · MATH 241 · Advance Calculus

Complex Calculus

Analytic Functions, Cauchy-Riemann, Complex Integration

S2. Complex Calculus

Roots & Geometry of Complex Numbers

Q1. Find all values of z for which $z^5 = -32$, and locate these values in the complex plane. **(10 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q2. Find the image of $\mathrm{Im}(z) \geq 0$ under the transformation $w = \dfrac{z-i}{iz-1}$. Sketch the region $\mathrm{Im}(z) \geq 0$ and its image. **(8 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q3. Show that $\mathrm{Log}(-1+i\sqrt{3})^2 \neq 2\,\mathrm{Log}(-1+i\sqrt{3})$. **(6 marks)** [MATH 241 | L-2/T-1 | 2022–23, 2016–17, 2015–16]

Q4. Find all roots of $(-8)^{1/6}$ in rectangular coordinates, exhibit them as vertices of a regular polygon, and identify the principal root. **(10 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q5. Identify $|z-3i|+|z+3i| = 8$ and $|z-1| = |z+i|$ geometrically. **(15 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q6. Let R be the region in the z-plane bounded by $x = 0$, $y = 0$, $x = 2$, $y = 1$. Determine the region R' of the w-plane into which R is mapped under $w = \sqrt{2}\,e^{i\pi/4}z + (1-2i)$. **(15 marks)** [MATH 245 | L-2/T-1 | 2019–20]

Q7. If $|z_1| = |z_2|$ and $\arg(z_1)+\arg(z_2) = 0$, show that z_1 and z_2 are complex conjugates. **(10 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q8. Prove that if the ratio $\dfrac{z-i}{z-1}$ is purely imaginary, then the point z lies on the circle whose centre is $\frac{1}{2}(1+i)$ and radius $\dfrac{1}{\sqrt{2}}$. **(10 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q9. Find all roots of $(-2\sqrt{3}-2i)^{1/4}$ in rectangular coordinates and exhibit them graphically. **(13 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q10. Describe mathematically and graphically the region represented by $z(\bar{z}+2) = 3$. **(10 marks)** [MATH 241 | L-2/T-1 | 2015–16, 2014–15]

Q11. Find the image of $\mathrm{Re}(z) \geq 0$ under the transformation $w = \dfrac{z-1}{z+1}$. Sketch the region $\mathrm{Re}(z) \geq 0$ and its image. **(12 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q12. Find the principal argument of $z = (\sqrt{3}-i)^6$. **(10 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q13. Prove that $|z_1+z_2| \geq \big||z_1|-|z_2|\big|$ where z_1, z_2 are nonzero complex numbers. **(8 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q14. Show that the equation $|z-z_0| = R$ represents a circle centred at z_0 with radius R, which can be written as $z\bar{z}-2\,\mathrm{Re}(z\bar{z}_0)+|z_0|^2 = R^2$. **(7 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q15. Find all the roots of $(-2\sqrt{3}-2i)^{1/4}$ in rectangular coordinates and exhibit the distinct roots graphically. **(10 marks)** [MATH 241 | L-2/T-1 | 2013–14, 2011–12]

Q16. Prove that $|z_1+z_2|^2+|z_1-z_2|^2 = 2|z_1|^2+2|z_2|^2$ and hence deduce that

$$\left|a+\sqrt{a^2-b^2}\right|+\left|a-\sqrt{a^2-b^2}\right| = |a+b|+|a-b|,$$

where z_1, z_2, a, b are complex numbers. **(20 marks)** [MATH 241 | L-2/T-1 | 2013–14]

Q17. Describe mathematically and graphically the region represented by $|z+2-3i|+|z-2+3i| < 10$. **(5 marks)** [MATH 241 | L-2/T-1 | 2013–14]

Q18. Find all the roots of the following equations in rectangular coordinates, exhibit them geometrically, and point out which is the principal root:

(a) $z^4+2\sqrt{3}+2i = 0$

(b) $z^3+1-i = 0$

(18 marks) [MATH 241 | L-2/T-1 | 2012–13]

Q19. Prove the triangle inequalities of complex numbers:

(a) $|z_1+z_2| \leq |z_1|+|z_2|$

(b) $|z_1-z_2| \geq \big||z_1|-|z_2|\big|$

(17 marks) [MATH 241 | L-2/T-1 | 2012–13]

Q20. Show that the equation $\left|z + \frac{1}{z}\right| = 2$ represents two circles with centres $(0, 1)$, $(0, -1)$ and radius $\sqrt{2}$. **(10 marks)** [MATH 241 | L-2/T-1 | 2011–12]

Limits and Continuity of Complex Functions

Q1. Test $f(z) = \begin{cases} (\bar{z})^2/z, & z \neq 0 \\ 0, & z = 0 \end{cases}$ for continuity and differentiability at $z = 0$. **(12/10 marks)** [MATH 241 | L-2/T-1 | 2022–23, 2014–15]

Q2. Evaluate the following limit:

$$\lim_{z \to i} \left[\frac{\tan^{-1}(z^2+1)^2}{\sin^2(z^2+1)} \right].$$

(10 marks) [MATH 245 | L-2/T-1 | 2021–22]

Analytic Functions & Cauchy-Riemann Equations

Q1. Derive the rectangular form of the Cauchy-Riemann equations and hence deduce the polar form. **(15 marks)** [MATH 241 | L-2/T-1 | 2011–12, 2013–14, 2014–15, 2023–24, 2024–25]

Q2. Show that $v(x, y) = \dfrac{x}{x^2 + y^2} + \cosh x \cos y$ is a harmonic function. Hence, find an analytic function $f(z) = u + iv$ and express $f(z)$ in terms of z. **(17 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q3. If $f(z) = e^{\bar{z}}$, apply the Cauchy-Riemann equations to show that $f'(z)$ does not exist at any point. **(8 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q4. Solve the complex equation $\cos z = 2$ by equating its real and imaginary parts. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q5. Define an analytic function with an illustrative example. Test the differentiability of the function $f(z) = |z|^2$ and show it is differentiable only at $z = 0$. **(13 marks)** [MATH 241 | L-2/T-1 | 2013–14, 2023–24]

Q6. Prove that the function $u = e^{-x}(x \sin y - y \cos y)$ is harmonic. Find its harmonic conjugate v such that $f(z) = u + iv$ is analytic, and express $f(z)$ explicitly in terms of z. **(15 marks)** [MATH 241 | L-2/T-1 | 2011–12, 2023–24]

Q7. Locate and name all the singularities of the function:

$$f(z) = \frac{z^8 + z^4 + 2}{(z-1)^3(3z+2)^2}.$$

Hence, determine the domain where $f(z)$ is analytic. **(8 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q8. For $f(z) = z^2 e^{2z}$, find the real component $u(x, y)$ and imaginary component $v(x, y)$ such that $f(z) = u + iv$. **(7 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q9. Show that $u(x, y) = \frac{1}{2}\log(x^2 + y^2)$ is harmonic in a specific domain. Find its harmonic conjugate $v(x, y)$ such that $f(z) = u + iv$ is analytic, and express $f(z)$ in terms of z. **(15 marks)** [MATH 241 | L-2/T-1 | 2013–14, 2022–23] [MATH 245 | L-2/T-1 | 2016–17]

Q10. Find the exact points where $f(z) = x^3 + i(1-y)^3$ is differentiable, and compute $f'(z)$ at those locations. **(6 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q11. Solve the complex equation $\cosh z = -2$ by equating its real and imaginary parts. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23, 2016–17, 2015–16, 2013–14]

Q12. Define a harmonic function and an analytic function. Prove that $u = \cos x \cosh y$ is harmonic, find its harmonic conjugate, and express the resulting analytic function $f(z)$ in terms of z. **(15 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q13. Evaluate the indefinite integral $\int \sqrt{1 + \sqrt{1+z}}\, dz$, explicitly stating the conditions under which the resulting expression remains valid. **(10 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q14. Test the differentiability of the function $f(z) = \dfrac{z^4}{64}$ within its natural domain. **(11 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q15. Prove that $u(x, y) = \sinh x \sin y + \dfrac{y}{x^2 + y^2}$ is harmonic, and compute its corresponding harmonic conjugate. **(15 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q16. Show that $|\cos z|^2 = (\cos x)^2 + (\sinh y)^2$ holds true for all $z = x + iy$. **(10 marks)** [MATH 245 | L-2/T-1 | 2019–20]

Q17. Determine which of the following functions $f(z)$ are entire. If a function is entire, compute its derivative $f'(z)$:

(a) $f(z) = \dfrac{1}{1 + |z - 1|^2}$

(b) $f(z) = 5^{(2^z)}$

(10 marks) [MATH 245 | L-2/T-1 | 2019–20]

Q18. Show that $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2 - c$ (where $c = 0$ or $c = 1$) is harmonic. Find $v(x, y)$ such that $f(z) = u + iv$ is analytic, and express $f(z)$ directly as a function of z. **(20 marks)** [MATH 241 | L-2/T-1 | 2012–13] [MATH 245 | L-2/T-1 | 2019–20]

Q19. If $f(z) = u(r, \theta) + iv(r, \theta)$, test the differentiability of the branch $f(z) = \sqrt{r}\, e^{i\theta/2}$ for $r > 0$ and $\alpha < \theta < \alpha + 2\pi$, and prove that $f'(z) = \dfrac{1}{2f(z)}$. **(15 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q20. Show that $u = e^x \cos y$ is a harmonic function. Find an analytic function $f(z) = u(x, y) + iv(x, y)$ and express $f(z)$ in terms of z. **(10 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q21. Let $f(z) = u + iv$ be differentiable at a non-zero point $z_0 = r_0 e^{i\theta_0}$. Prove that:

$$f'(z_0) = e^{-i\theta}(u_r + iv_r) = -\frac{i}{z_0}(u_\theta + iv_\theta),$$

where the partial derivatives are evaluated at (r_0, θ_0). **(18 marks)** [MATH 245 | L-2/T-1 | 2016–17]

Q22. Prove that $u(x, y) = e^x(x \cos y - y \sin y)$ is a harmonic function. Find its harmonic conjugate $v(x, y)$ and express $f(z) = u + iv$ as a function of z. **(15 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q23. If $f(z)$ is an analytic function of z, prove the identity:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2.$$

(10 marks) [MATH 241 | L-2/T-1 | 2011–12]

Q24. Find the polar form of the Cauchy-Riemann equations, and use the result to derive the Laplace equation in polar coordinates. **(15 marks)** [MATH 241 | L-2/T-1 | 2011–12]

Elementary Complex Functions

Q1. Find all roots of the equation $\cos z = 4$ by equating the real and imaginary parts in the equation. **(12 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q2. Show that $\mathrm{Log}(-1 + i)^2 \neq 2 \log(-1 + i)$. **(10 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q3. Show that Euler's formula $e^{i\theta} = \cos\theta + i \sin\theta$ continues to hold when θ is replaced by z. **(5 marks)** [MATH 241 | L-2/T-1 | 2014–15] [MATH 245 | L-2/T-1 | 2016–17]

Q4. Solve the equation $\sin z = 2$ by equating the real and imaginary parts in the equation. **(10 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q5. Express $\sinh z$ in the complex form $a + ib$ and show that $|\sinh z|^2 = \sinh^2 x + \sin^2 y$. **(10 marks)** [MATH 241 | L-2/T-1 | 2011–12]

Line Integral of Complex Functions

Q1. Evaluate $\displaystyle\int_C \bar{z}\, dz$ from $z = 0$ to $z = 4 + 2i$ along:

(a) $z = t^2 + it$

(b) the line from $z = 0$ to $z = 2i$ and then from $z = 2i$ to $z = 4 + 2i$

(16 marks) [MATH 241 | L-2/T-1 | 2012–13, 2015–16, 2024–25]

Q2. Prove that $\left|\int_C \frac{dz}{z^2-1}\right| \le \frac{\pi}{3}$ where C is the arc of $|z| = 2$ from $z = 2$ to $z = 2i$ in the first quadrant. **(10 marks)** [MATH 241 | L-2/T-1 | 2012–13, 2024–25]

Q3. Show that $\int_C \frac{dz}{z} = -\pi i$ or πi according as C is the semi-circular arc of $|z| = 1$ above or below the x-axis. **(5 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q4. Evaluate $\int_C (3xy + iy^2)\, dz$ along the straight line joining $z = i$ and $z = 2 - i$. **(12 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q5. Evaluate $\int_i^{2+5i} (3x - y)\, dx + (2y + x)\, dy$ along:

(a) the curve $y = x^2 + 1$

(b) straight lines from $(0, 1)$ to $(2, 1)$ and then from $(2, 1)$ to $(2, 5)$

(18 marks) [MATH 245 | L-2/T-1 | 2019–20]

Q6. If C is the boundary of the triangle with vertices 0, $3i$ and -4 (counterclockwise), show that $\left|\oint_C (e^z - \bar{z})\, dz\right| \le 57$. **(18 marks)** [MATH 245 | L-2/T-1 | 2019–20]

Q7. Let C be the arc of the circle $|z| = 2$ from $z = 2$ to $z = 2i$ that lies in the first quadrant. Show that

$$\left|\int_C \frac{z+4}{z^3-1}\, dz\right| \le \frac{6\pi}{7}.$$

(10 marks) [MATH 245 | L-2/T-1 | 2020–21]

Q8. Show that the integral of $\bar{z}$ along a semicircular arc $|z| = 1$ from -1 to 1 has the value $-\pi i$ or πi according as the arc lies above or below the real axis. **(10 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q9. Evaluate $\oint_C \bar{z}\, dz$ around (i) the circle $|z - 2| = 3$, and (ii) the ellipse $|z - 3| + |z + 3| = 10$. **(12 marks)** [MATH 245 | L-2/T-1 | 2016–17, 2013–14]

Q10. Evaluate $\int_C (z^2 + 3z)\, dz$ along the circle $|z| = 2$ from $(2, 0)$ to $(0, 2)$. **(10 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q11. If $f(z)$ is integrable along a curve C having finite length L and if there exists a positive number M such that $|f(z)| \le M$ on C, show that $\left|\oint_C f(z)\, dz\right| \le ML$. Use this result to show that $\left|\oint_C \frac{dz}{z^2-1}\right| \le \frac{\pi}{3}$, where C is the contour $|z| = 2$ in the first quadrant. **(17 marks)** [MATH 241 | L-2/T-1 | 2012–13]

Cauchy's Integral Formula & Theorem

Q1. Use Cauchy's Integral formula to evaluate $\oint_C \frac{z\, dz}{(9 - z^2)(z + i)}$ where $C\colon |z| = 2$, taken in the positive sense. **(12 marks)** [MATH 245 | L-2/T-1 | 2016–17] [MATH 241 | L-2/T-1 | 2024–25]

Q2. Use Cauchy's integral formula to show that $\frac{1}{2\pi i}\oint_C \frac{e^{zt}}{(z^2+1)^2}\, dz = \frac{1}{2}(\sin t - t\cos t)$ if $t > 0$ and $C\colon |z| = \frac{3}{2}$. **(15 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q3. State Cauchy-Goursat theorem and apply it to find $\int_C \operatorname{sech} z\, dz$ where C is the circle $|z| = 1$. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q4. Use Cauchy's integral formula to evaluate $\oint_C \frac{4 - 3z}{z(z-1)(z-2)}\, dz$ where $C\colon |z| = \frac{3}{2}$. **(13 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q5. Compute $\oint_C \frac{7 - 2z}{(z+1)(z-1)(z-3)}\, dz$ over $C\colon |z| = \frac{5}{2}$ using Cauchy's Integral Formula. **(10 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q6. State and prove Cauchy's Integral formula. **(12 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q7. Use Cauchy's Integral formula to evaluate $\oint_C \frac{\cos z}{z^2(z^2+8)}\,dz$ where $C\colon |z| = 2$. **(7 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q8. Use Cauchy's integral formula to evaluate the integral

$$\frac{1}{2\pi i}\oint_C \frac{e^{-3z}}{z^2+4}\,dz,$$

where C is the circle $|z| = 3$, taken in the positive sense. **(10 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q9. Use Cauchy's integral formula to evaluate $\oint_C \frac{\sin^6 z}{(z-\pi/6)^3}\,dz$, where C is the circle $|z| = 1$. **(10 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q10. Use Cauchy's integral formula to evaluate $\oint_C \frac{z}{(9-z^2)(z+i)}\,dz$, where C is the circle $|z| = 2$, taken in the positive sense. **(12 marks)** [MATH 245 | L-2/T-1 | 2016–17]

Q11. Using Cauchy's integral formula, find the value of $\oint_C \frac{dz}{(z^2+4)^2}$ around the circle $C\colon |z-i| = 2$, taken in the positive sense. **(10 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q12. Evaluate $\oint_C \frac{1}{(z+2)(z-4i)}\,dz$ by Cauchy's integral formula around the circle $C\colon |z-i| = 4$ taken in the positive sense. **(8 marks)** [MATH 241 | L-2/T-1 | 2013–14]

Q13. Use Cauchy's integral formula to evaluate

$$\oint_C \frac{\cos z}{z(z^2+8)}\,dz,$$

where C is the boundary of the square defined by the lines $x = \pm 2$, described in the positive sense. **(17 marks)** [MATH 241 | L-2/T-1 | 2012–13]

Q14. State and prove Cauchy's integral formula. Use this formula to evaluate

$$\oint_C \frac{z}{z^2+1}\,dz,$$

where C is the circle $|z+i| = 1$. **(18 marks)** [MATH 241 | L-2/T-1 | 2011–12]

Q15. Evaluate

$$\oint_C \frac{z^4+1}{z^2(z+2)(2z+1)}\,dz,$$

where C is the circle $|z| = 1$. **(17 marks)** [MATH 241 | L-2/T-1 | 2011–12]

Taylor Series

Q1. Expand $f(z) = \frac{z+1}{z-2}$ in a Taylor series in powers of z and state the region where the expansion is valid. **(8 marks)** [MATH 245 | L-2/T-1 | 2016–17]

Laurent Series

Q1. Expand $f(z) = \frac{z}{(z-1)(2-z)}$ in a Laurent series valid for $0 < |z-2| < 1$. **(10 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q2. Expand $f(z) = \frac{5}{(z+1)(z+3)}$ in a Laurent series valid for

(a) $1 < |z| < 3$

(b) $|z| < 1$

(10 marks) [MATH 245 | L-2/T-1 | 2020–21]

Q3. Find the Laurent series of $f(z) = \dfrac{z^2}{z^2 - 3z + 2}$ in each domain: (i) $1 < |z| < 2$; (ii) $1 < |z-3| < 2$. **(15 marks)** [MATH 245 | L-2/T-1 | 2019–20]

Q4. Express $f(z) = \dfrac{1}{(z^2+1)(z+2)}$ in a Laurent series valid in the region $1 < |z| < 2$. **(13 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q5. State Laurent's theorem. Find two Laurent series of $f(z) = \dfrac{1}{(z+1)(z+3)}$ valid in the regions (i) $1 < |z| < 3$ and (ii) $0 < |z+1| < 2$. **(15 marks)** [MATH 245 | L-2/T-1 | 2016–17]

Q6. State Laurent's theorem. Expand $f(z) = \dfrac{1}{(z-1)(2-z)}$ in a Laurent series valid for (i) $1 < |z| < 2$ and (ii) $0 < |z-2| < 1$. **(18 marks)** [MATH 241 | L-2/T-1 | 2014–15, 2011–12]

Q7. Find a Laurent series expansion for $f(z) = \dfrac{1}{(z+1)(z-3i)}$ in powers of $(z+1)$ and state the region of convergence of the series. **(8 marks)** [MATH 241 | L-2/T-1 | 2013–14]

Residues & Residue Theorem

Q1. Compute: **(i)** $\displaystyle\oint_C \frac{\coth z}{(z-1)(z-i)}\,dz$ where $C : |z| = 2$, **(ii)** $\displaystyle\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)}\,dz$ where $C : |z| = 3$. **(15 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q2. Determine the poles of $\dfrac{3+4z}{z(z+1)^2(z-3)}$ over $C : |z| = 4$ and the residue at each pole. Hence find $\displaystyle\oint_C \frac{3+4z}{z(z+1)^2(z-3)}\,dz$. **(15 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q3. Evaluate $\displaystyle\oint_C \frac{\sin \pi z}{(z-3)^3}\,dz$, where C is the circle $|z| = 4$, by Cauchy's residue theorem. **(15 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q4. Evaluate $\displaystyle\oint_C \frac{e^{-iz}}{(z+3)(z-i)^2}\,dz$ by Cauchy's Residue theorem, where $C = \{z : z = 1 + 2e^{i\theta},\ 0 \le \theta \le 2\pi\}$. **(17 marks)** [MATH 245 | L-2/T-1 | 2019–20]

Q5. Evaluate $\displaystyle\int_0^{\pi} \frac{d\theta}{2 - \cos\theta}$ by the method of contour integration. **(17 marks)** [MATH 245 | L-2/T-1 | 2019–20]

Q6. Evaluate $\displaystyle\oint_C \frac{1}{z(2z-1)(z-4)}\,dz$, where $C = \{z : |z+2| + |z-2| = 6,\ \text{positively oriented}\}$, by Cauchy's residue theorem. **(12 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q7. Evaluate by Cauchy's residue theorem where C is the circle $|z| = 3$:

(a) $\displaystyle\oint_C \frac{1+z^2}{(z-1)^2(z+2i)}\,dz$

(b) $\displaystyle\oint_C \frac{\sin z}{z^2(z^2+4)}\,dz$

(c) $\displaystyle\oint_C \frac{\sinh z}{z^4}\,dz$

(23 marks) [MATH 245 | L-2/T-1 | 2016–17]

Q8. What is a residue? Expand $f(z) = \dfrac{1}{(z+1)(z+3)}$ in a Laurent series valid for (i) $|z| > 3$; (ii) $0 < |z+1| < 2$. Write the residue in each case. **(17 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q9. Evaluate $\displaystyle\frac{1}{2\pi i}\oint_C \frac{e^{zt}}{z^2(z^2+2z+2)}\,dz$ around the circle $C : |z| = 3$. **(18 marks)** [MATH 241 | L-2/T-1 | 2015–16]

Q10. Evaluate the following integrals using residues and contours:

(a) $\displaystyle\int_{-\infty}^{+\infty} \frac{x^2}{(x^2+1)^2(x^2+2x+2)}\,dx$ **(17 marks)**

(b) $\displaystyle\int_0^{\infty} \frac{x^3 \sin x}{(x^2+1)(x^2+9)}\,dx$ **(18 marks)**

[MATH 241 | L-2/T-1 | 2015–16]

Q11. State Cauchy's residue theorem and use it to find $\oint_C \frac{3z^3+2}{(z-1)(z^2+9)}\,dz$ counterclockwise around:

(a) $|z-2| = 2$

(b) $|z| = 4$

(17 marks) **[MATH 241 | L-2/T-1 | 2014–15]**

Q12. Find the residue of $f(z) = \frac{(\log z)^3}{z^2+1}$ at the point $z = i$. **(8 marks)** **[MATH 241 | L-2/T-1 | 2013–14]**

Q13. Use Cauchy's residue theorem to evaluate $\oint_C \frac{1+z^2}{(z-1)^2(z+2i)}\,dz$, where C is the circle $|z| = 3$. **(11 marks)** **[MATH 241 | L-2/T-1 | 2013–14]**

Q14. Determine the poles of $f(z) = \frac{e^z}{z^2+\pi^2}$ and find residues at each pole. Then use the residue theorem to evaluate

$$\oint_C \frac{e^z}{z^2+\pi^2}\,dz,$$

where C is the circle $|z| = 4$. **(18 marks)** **[MATH 241 | L-2/T-1 | 2012–13]**

Q15. Evaluate: (a) $\int_0^\infty \frac{dx}{x^4+1}$ (b) $\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta}\,d\theta$ using residues and contours. **(35 marks)** **[MATH 241 | L-2/T-1 |]**

Contour Integration

Type 1: Integrals of Rational Trigonometric Functions

Q1. Evaluate by contour integration: $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\,d\theta$. **(17 marks)** **[MATH 241 | L-2/T-1 | 2012–13]** **[MATH 245 | L-2/T-1 | 2021–22]**

Q2. Evaluate the following integral using the method of contour integration:

$$\int_0^{2\pi} \frac{d\theta}{(5-3\cos\theta)^2}.$$

(17 marks) **[MATH 245 | L-2/T-1 | 2020–21]**

Q3. Evaluate by contour integration: $\int_0^{2\pi} \frac{\sin 2\theta}{5-3\cos\theta}\,d\theta$. **(17 marks)** **[MATH 245 | L-2/T-1 | 2018–19]**

Q4. Evaluate by contour integration: $\int_0^{2\pi} \frac{\cos(3\theta)}{5+4\cos\theta}\,d\theta$. **(17 marks)** **[MATH 245 | L-2/T-1 | 2016–17]**

Q5. Evaluate by contour integration: $\int_0^{2\pi} \frac{\cos 2\theta}{1-2a\cos\theta+a^2}\,d\theta, \quad a^2 < 1$. **(18 marks)** **[MATH 241 | L-2/T-1 | 2014–15]**

Q6. Evaluate by contour integration: $\int_0^{2\pi} \frac{d\theta}{1+a\cos\theta}, \quad -1 < a < 1$. **(18 marks)** **[MATH 241 | L-2/T-1 | 2013–14]**

Type 2: Rational Algebraic Functions

Q7. Evaluate by contour integration: $\int_{-\infty}^{\infty} \frac{dx}{1+x^6}$. **(18 marks)** **[MATH 245 | L-2/T-1 | 2021–22]**

Q8. Evaluate the following integral using the method of contour integration:

$$\int_{-\infty}^{\infty} \frac{x^2\,dx}{1+x^6} \quad \left(\text{or } \int_0^\infty \frac{x^2\,dx}{1+x^6}\right).$$

(18 marks) **[MATH 241 | L-2/T-1 | 2013–14]** **[MATH 245 | L-2/T-1 | 2020–21]**

Q9. Evaluate by contour integration: $\int_0^\infty \frac{dx}{(1+x^2)^2} \quad \left(\text{or } \int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)^2}\,dx\right)$. **(18 marks)** **[MATH 241 | L-2/T-1 | 2012–13]** **[MATH 245 | L-2/T-1 | 2018–19]**

Q10. Evaluate by contour integration: $\int_0^\infty \frac{dx}{(a^2+x^2)^2}$. **(18 marks)** **[MATH 245 | L-2/T-1 | 2016–17]**

Q11. Evaluate by contour integration:

$$\int_0^\infty \frac{x^2 dx}{(x^2+9)(x^2+4)^2}.$$

(17 marks) [MATH 241 | L-2/T-1 | 2011–12]

Type 3: Fourier-Type Integrals (Improper Integrals with Sine/Cosine)

Q12. Evaluate by contour integration: $\displaystyle\int_0^\infty \frac{\cos ax}{x^2+b^2}\,dx, \quad a>0,\ b>0.$ **(17 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q13. Evaluate by contour integration:

$$\int_{-\infty}^\infty \frac{x\sin \pi x}{x^2+2x+5}\,dx.$$

Conformal Mapping

Type 1: Geometric Images under Mapping

Q1. What is the region of the w-plane into which the rectangular region in the z-plane bounded by $x=0$, $y=0$, $x=2$, $y=1$ is mapped under $w=z+(2-i)$? **(10 marks)** [MATH 245 | L-2/T-1 | 2021–22]

Q2. Transform the circle $x^2+y^2-25x=0$ into a straight line under the transformation $w=\dfrac{5z+4}{z+3}$. **(12 marks)** [MATH 245 | L-2/T-1 | 2020–21]

Q3. Show that the transformation $w=\dfrac{1+iz}{z+i}$ transforms the real axis on the z-plane onto a circle in the w-plane. Find the centre and radius of this circle. **(15 marks)** [MATH 245 | L-2/T-1 | 2018–19]

Q4. Find the image of the hyperbola $x^2-y^2=1$ under the transformation $w=\dfrac{1}{z}$ and sketch the graph of the image. **(12 marks)** [MATH 241 | L-2/T-1 | 2013–14]

Q5. Show that the transformation $w=\dfrac{5-4z}{z-2}$ transforms the circle $|z|=1$ into a circle in the w-plane. Find the centre and radius of this circle. **(17 marks)** [MATH 241 | L-2/T-1 | 2011–12]

Type 2: Finding Bilinear Transformations

Q5. Find the bilinear transformation which maps $z_1=-i$, $z_2=0$, $z_3=i$ into $w_1=-1$, $w_2=i$, $w_3=1$. Hence find the image of the y-axis. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q6. Find the bilinear transformation which maps the points $z_1=-i$, $z_2=0$, $z_3=i$ to $w_1=-1$, $w_2=i$, $w_3=1$ respectively. **(12 marks)** [MATH 245 | L-2/T-1 | 2016–17]

Q7. Find the bilinear transformation which maps the points $z_1=2$, $z_2=i$, $z_3=-2$ to $w_1=1$, $w_2=i$, $w_3=-1$ respectively. **(10 marks)** [MATH 241 | L-2/T-1 | 2014–15]

Q8. Find a bilinear transformation which maps the upper half of the z-plane into the unit circle in the w-plane such that $z=i$ is mapped to $w=0$ and the point at infinity is mapped to $w=-1$. **(17 marks)** [MATH 241 | L-2/T-1 | 2012–13]

BUET · MATH 241 · Advance Calculus

Partial Differential Equations

Formation, First & Second Order PDE, Separation of Variables

S3. Partial Differential Equations

Introduction & Formation of PDE

Q1. Form a PDE by eliminating the arbitrary functions f and g from $z = f(x^2 - y) + g(x^2 + y)$. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q2. Form a PDE by eliminating the arbitrary function f from $z = e^{mx} f(x + y)$. **(12 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q3. Form a PDE by eliminating the arbitrary constants from $z = ax + by + a^2 + b^2$. **(12 marks)** [MATH 147 | L-1/T-2 | 2016–17]

First Order Linear PDE

Q1. Solve $(1 + y)p + (1 + x)q = z$ using Lagrange's method. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Find the surface satisfying $4yzp + q + 2y = 0$ and passing through $y^2 + z^2 = 1$, $x + z = 2$. **(13 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q3. Apply Lagrange's method to solve $(xy^3 - 2x^4)p + (2y^4 - x^3y)q = 9z(x^3 - y^3)$. **(11 marks)** [MATH 143 | L-1/T-2 | 2013–14] [MATH 241 | L-2/T-1 | 2023–24]

Q4. Find the integral surface of $(y + zx)p + (x + yz)q = z^2 - 1$ which passes through $x = t$, $y = 1$, $z = t^2$. **(12 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q5. Find the integral surface satisfying $(2xy - 1)p + (z - 2x^2)q = 2(x - yz)$ and passing through $x = 1$, $y = 0$. **(12 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q6. Solve: $(y - zx)p + (x + yz)q = x^2 + y^2$. **(15 marks)** [MATH 147 | L-1/T-2 | 2016–17, 2020–21]

Q7. Find the integral surface of $2y(z - 3)p + (2x - z)q = y(2x - 3)$ which passes through the circle $z = 0$, $x^2 + y^2 = 2x$. **(20 marks)** [MATH 143 | L-1/T-2 | 2010–11] [MATH 147 | L-1/T-2 | 2016–17, 2019–20, 2020–21]

Q8. Find the integral surface of $(2xy + 1)p - (z + 2x^2)q = 2(x - yz)$ which passes through the line $x = 1$, $y = 0$. **(20 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q9. Find the integral surface of $(x - y)p + (y - x - z)q = z$ which passes through $x^2 + y^2 = 1$, $z = 1$. **(17 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q10. Solve: $(x^2 - y^2 - yz)p + (x^2 - y^2 - zx)q = z(x - y)$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q11. Find the integral surface of $(x - y)p + (y - z - x)q = z$ through the circle $z = 1$, $x^2 + y^2 = 1$. **(16 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q12. Solve: $x(x + y)p - y(x + y)q + (x - y)(2x + 2y + z) = 0$. **(15 marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q13. Find the integral surface of $px^2 + qy^2 + z^2 = 0$ which passes through the hyperbola $xy = x + y$, $z = 1$. **(16.5 marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q14. Find the integral surface of the first order linear partial differential equation:

$$x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$$

which contains the line $x + y = 0$, $z = 1$. **(15 marks)**

Q15. Solve: $pz - qz = z^2 + (x + y)^2$. **(15 marks)** [MATH 143 | L-1/T-2 | 2012–13]

Q16. Solve: $p\cos(x + y) + q\sin(x + y) = z$. **(15 marks)** [MATH 143 | L-1/T-2 | 2011–12]

First Order Non-linear PDE

Q1. Find a complete integral of $px + qy = pq$ using Charpit's method. **(12 marks)** [MATH 143 | L-1/T-2 | 2012–13] [MATH 241 | L-2/T-1 | 2024–25]

Q2. Use Charpit's method to find a complete integral and singular integral (if exists) of $2(z + px + qy) = yp^2$. **(12 marks)** [MATH 143 | L-1/T-2 | 2010–11] [MATH 241 | L-2/T-1 | 2023–24]

Q3. Find the complete integral of $2x(z^2q^2 + 1) = pz$. **(13 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q4. Find the complete, singular, and general integrals of $(p^2 + q^2)y = qz$. **(16 marks)** [MATH 143 | L-1/T-2 | 2011–12] [MATH 147 | L-1/T-2 | 2016–17, 2020–21]

Q5. Find the complete integral and singular integral (if it exists) of $2xz - px^2 - 2qxy + pq = 0$. **(26 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q6. Find the complete integral and singular integral (if exist) of $p - 3x^2 = q^2 - y$. **(20 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q7. Find the complete, singular (if exists) and general integral of $z^2(p^2z^2 + q^2) = 1$. **(17 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q8. Find the complete and singular integral (if exists) of $z = px + qy + c\sqrt{1 + p^2 + q^2}$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q9. Solve: $(p^2 + q^2)x = pz$. **(15 marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q10. Find the complete integral and singular integral (if it exists) of $z^2(p^2 + q^2 + 1) = 4$. **(15 marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q11. Solve: $p(q^2 + 1) + (3 - z)q = 0$. **(16.5 marks)** [MATH 143 | L-1/T-2 | 2012–13]

Q12. Solve: $p^2 + q^2 - 2px - 2qy + 1 = 0$. **(17.5 marks)** [MATH 143 | L-1/T-2 | 2011–12]

Second Order Linear PDE: Classifications

Q1. Classify and reduce to canonical form the partial differential equation:

$$u_{xx} + 2\sin x\, u_{xy} - \cos^2 x\, u_{yy} - \cos x\, u_y = 0.$$

(20 marks) [MATH 147 | L-1/T-2 | 2015–16]

Homogenous linear PDE with constant coefficients

Q1. Solve $(D_x^2 + 2D_xD_y + D_y^2)z = \sin(2x + 3y)$. **(10 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Solve $(6D_x^2 - D_xD_y - 12D_y^2)z = e^{2x+3y} + x^2y^2$. **(11 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q3. Solve $(D_x^3 - 2D_x^2D_y - D_xD_y^2 + 2D_y^3)z = f(x, y)$ where:

(a) $f(x, y) = \sin(2x + y)$ **(11 marks)** [MATH 241 | L-2/T-1 | 2022–23]

(b) $f(x, y) = e^{x+y}$ **(15 marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q4. Solve the following homogeneous partial differential equation: $(D_x^2 - 2D_xD_y)z = 4e^{2x} + 5x^2y$ **(13 marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q5. Solve the following: $(D_x^2 + 4D_xD_y + 4D_y^2)z = e^{2x+y}$ **(13 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q6. Solve the following equations: $(D_x^2 - D_xD_y - 2D_y^2)z = (y - 1)e^x$ **(15 marks)** [MATH 143 | L-1/T-2 | 2014–15] [MATH 147 | L-1/T-2 | 2016–17]

Q7. Solve the homogeneous equation: $(D_x^2 + D_xD_y - 2D_y^2)z = (2x + y)^{1/2} + e^{y+2x}$. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Q8. Solve: $(D_x^3 - 4D_x^2D_y + 4D_xD_y^2)z = 4\sin(2x + y)$. **(15 marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q9. Solve: $(6D_x^2 + 8D_xD_y - 8D_y^2)z = e^{2x+y} \cdot xy^2$. **(14 marks)** [MATH 143 | L-1/T-2 | 2012–13]

Q10. Solve: $(3D_x^2 + 7D_xD_y + 2D_y^2)z = e^{x+3y} \cdot x^2y^2$. **(17.5 marks)** [MATH 143 | L-1/T-2 | 2011–12]

Q11. Solve: $(4D_x^2 - 4D_xD_y + D_y^2)z = \log(x+2y) + e^{3x-2y}$. **(15 marks)** [MATH 143 | L-1/T-2 | 2010–11]

Non-homogeneous linear PDE with constant coefficients

Q1. Solve $(D_x^2 - D_y)(D_x - 2D_y)z = e^{2x+y} + xy$. **(13 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Solve $(3D_x^2 - 2D_y^2 + D_x - 1)z = 4e^{x+y}\sin(x+y)$. **(12 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q3. Solve $(D_x^2 - 4D_xD_y + 4D_y^2 + D_x - 2D_y)z = e^{x+y} + xy$. **(12 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q4. Solve $(D_x^2D_y + D_y^2 - 2)z = 4e^{x+y}\cos(x+y)$. **(12 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q5. Solve the following non-homogeneous partial differential equation:

(a) $(D_x^2 + 2D_xD_y + D_y^2 - 2D_x - 2D_y)z = \cos(x+2y)$ **(13 marks)**

[MATH 147 | L-1/T-2 | 2020–21]

Q6. Find the general solution of: $(D_x^2 + 3D_y - 2D_x)z = (x^2 + 2y^2)e^{2x+y}$. **(22 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q7. Solve: $(D_x^2 - D_y^2 - 3D_x + 3D_y)z = xy + e^{x+2y}$. **(15 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q8. Solve the following:

(a) $(D_x^2 - D_xD_y + D_y - 1)z = \sin(x+2y) + e^x$ **(13 marks)**

[MATH 147 | L-1/T-2 | 2017–18]

Q9. Solve the following equations:

(a) $(D_x^2 - D_xD_y + D_y - 1)z = \cos(x+2y) + e^y + xy + 1$ **(10 marks)** [MATH 147 | L-1/T-2 | 2016–17]

(b) $(D_x^2 + D_xD_y + D_y - 1)z = \cos(x+2y)$ **(15 marks)** [MATH 143 | L-1/T-2 | 2010–11]

Q10. Solve: $(D_x^2 - D_xD_y + 2D_y^2 + 2D_x + 2D_y)z = e^{3x+4y} + xy$. **(16.5 marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q11. Solve: $(3D_x - 4D_y + 2)(5D_x + 3D_y - 6)z = \sin(2x - 3y)$. **(15 marks)** [MATH 143 | L-1/T-2 | 2012–13]

Q12. Solve: $(2D_x - 5D_y + 6)(3D_x + 2D_y - 3)z = \cos(2x + 3y)$. **(15 marks)** [MATH 143 | L-1/T-2 | 2011–12]

Euler-Cauchy type PDE

Q1. Solve the Euler-Cauchy partial differential equation: $(x^2D_x^2 - 4xyD_xD_y + 4y^2D_y^2 + 6yD_y)z = x^3y^4$. **(12 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Solve $(x^2D_x^2 - xyD_xD_y - 2y^2D_y^2 + xD_x - 2yD_y)z = x^2y^3$. **(12 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q3. Reduce $(x^2D_x^2 - 2xyD_xD_y - 3y^2D_y^2 + xD_x - 3yD_y)z = 0$ to a linear PDE with constant coefficients and hence solve. **(10 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q4. Solve the variable-coefficient equation $(x^2D_x^2 - y^2D_y^2 + xD_x - yD_y)z = \ln x$. **(14 marks)** [MATH 143 | L-1/T-2 | 2011–12] [MATH 147 | L-1/T-2 | 2016–17, 2015–16]

Q5. Solve: $(x^2D_x^2 - 2xyD_xD_y + y^2D_y^2 - xD_x + 3yD_y)z = 8y/x$. **(16.5 marks)** [MATH 143 | L-1/T-2 | 2014–15]

Q6. Solve: $(x^2D_x^2 - 3xyD_xD_y + 2y^2D_y^2 + xD_x + 2yD_y)z = x + 2y$. **(15 marks)** [MATH 143 | L-1/T-2 | 2013–14]

Q7. Solve the variable-coefficient problem $(x^2D_x^2 - xyD_xD_y - 2y^2D_y^2 + xD_x - 2yD_y)z = f(x,y)$ where:

(a) $f(x,y) = \log(y/x) - \frac{1}{2}$ **(17.5 marks)** [MATH 143 | L-1/T-2 | 2012–13]

(b) $f(x,y) = \log\left(\frac{y}{x}\right)$ **(16.5 marks)** [MATH 143 | L-1/T-2 | 2010–11]

Solution by Separation of Variables

Q1. Solve $\dfrac{\partial u}{\partial x} = 4\dfrac{\partial u}{\partial y}$ with $u(0, y) = 8e^{-3y}$ using the method of separation of variables. **(15 marks)** [MATH 147 | L-1/T-2 | 2015–16]

Boundary Value Problems

Q1. A string is fastened at two points l apart. Motion is started by displacing the string into $y = k(lx - x^2)$ and releasing it at $t = 0$. Find the displacement of any point at distance x from one end at time t. **(18 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q2. Use the separation of variables method to solve the related partial differential equations and find the temperature distribution in a bar of length 2 whose ends are kept at zero, the lateral surface is insulated, and the initial temperature is given by $\sin\dfrac{\pi x}{2} + 3\sin\dfrac{5\pi x}{2}$. **(17 marks)** [MATH 241 | L-2/T-1 | 2024–25]

Q3. Solve the wave equation $\dfrac{\partial^2 u}{\partial t^2} = 16\dfrac{\partial^2 u}{\partial x^2}$, $0 < x < 2$, $t > 0$, subject to the boundary and initial conditions:

$$u(0, t) = 0, \quad u(2, t) = 0; \quad u(x, 0) = 6\sin\pi x - 3\sin 4\pi x, \quad \frac{\partial u}{\partial t}(x, 0) = 0.$$

Find the explicit expression for $u(x, t)$. **(18 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q4. A bar 10 cm long has its ends A and B maintained at 50°C and 100°C respectively until a steady state prevails. The temperature at A is then suddenly raised to 90°C and at B lowered to 60°C. Find the temperature distribution in the bar at any subsequent time t. **(17 marks)** [MATH 241 | L-2/T-1 | 2023–24]

Q5. Solve the wave equation $\dfrac{\partial^2 u}{\partial t^2} = 16\dfrac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(0, t) = u(2, t) = 0$, $u(x, 0) = x(2 - x)$, and $\left.\dfrac{\partial u}{\partial t}\right|_{t=0} = 0$ for $0 < x < 2$. **(25 marks)** [MATH 241 | L-2/T-1 | 2022–23]

Q6. Solve the wave equation $\dfrac{\partial^2 y}{\partial t^2} = 4\dfrac{\partial^2 y}{\partial x^2}$ by separation of variables satisfying the conditions: $y(0, t) = 0$, $y(\pi, t) = 0$ for $t \geq 0$; $y(x, 0) = \pi x - x^2$ and $y_t(x, 0) = 0$ for $0 \leq x \leq \pi$. **(20.67 marks)** [MATH 147 | L-1/T-2 | 2020–21]

Q7. Solve the following boundary value problem for heat conduction:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(4, t) = 0, \quad u(x, 0) = 6\sin\frac{\pi x}{2} + 3\sin\pi x,$$

where $0 < x < 4$ and $t > 0$. **(24 marks)** [MATH 147 | L-1/T-2 | 2019–20]

Q8. Solve the heat conduction equation:

$$\frac{\partial u}{\partial t} = 100\frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(1, t) = 0, \quad u(x, 0) = \sin 2\pi x - \sin 5\pi x,$$

where $0 < x < 1$ and $t > 0$. **(25 marks)** [MATH 147 | L-1/T-2 | 2018–19]

Q9. Solve the boundary value problem:

$$4\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(2, t) = 0, \quad u(x, 0) = 2\sin\frac{\pi x}{2} - \sin\pi x + 4\sin 2\pi x,$$

where $0 < x < 2$ and $t > 0$. **(20 marks)** [MATH 147 | L-1/T-2 | 2017–18]

Q10. Solve the boundary value problem and give a physical interpretation of the solution:

$$\frac{\partial u}{\partial t} = 3\frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(2, t) = 0, \quad u(x, 0) = x,$$

where $0 < x < 2$ and $t > 0$. **(16 marks)** [MATH 147 | L-1/T-2 | 2016–17]

Q11. Solve the boundary value problem by separation of variables:

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(\pi, t) = 0, \quad u(x, 0) = 2\sin 3x - 4\sin 5x,$$

where $0 < x < \pi$ and $t > 0$. **(20 marks)** [MATH 147 | L-1/T-2 | 2015–16]